Technical Requirements

Software

Area	Technology
Frontend	ReactJS (Vite dev server on port 5173)
Backend	Python, FastAPI, Uvicorn
Computer vision	OpenCV, PyTorch, torchvision
LLM	Meta Llama served through Ollama
Audio	Voice module in `backend/voice.py`

Hardware

- Laptop or PC with enough RAM to run the chosen Ollama model (smaller models such as 3B size run on low-RAM machines).
- GPU optional. CPU works but is slower.

Setup

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
cd frontend && npm install
```

Data requirements

- Food image dataset organized as one folder per dish class.
- Custom images of your own (`own_train.zip`) for dishes missing from the base dataset.
- Nutrition lookup per dish class (`backend/nutrition.py`).
- Class lists: `models/classes.json` (food), `models/freshness_classes.json` and `models/freshness_binary_classes.json` (freshness).
- Own test photos for freshness: `data/own_test` (27 images: 14 fresh, 13 rotten).